

“Computational Intelligence” – Exercise 1

Task 1: Preparation

- a) Run the Visual Studio Code (VSCode ) application. Under “File”=>”Open Folder” select a folder where you want to save your code (recommended: “M:\”, you can create a folder “KDD” there and open it in VSCode).
- b) Next, click on “File”=>”New File...” and select “Jupyter Notebook”. Write in the first cell **`print('Hello World!')`** and run it by clicking on “Execute cell” . Now you are prompted to select a kernel, click on “Python Environments” and select “Tensorflow-Env (Python 3.10.11)” which should be already recommended.
- c) Now execute your first cell, it should print “Hello World!” directly below it.

If this is done, you should be good to go for the rest of the exercise as well as the upcoming exercises in the future. Feel free to familiarize yourself with the VSCode application and the Jupyter Notebook environment. Ask me if in doubt.

Task 2

- Create a list **x** with values **5, 10** and **15**.
- Print it.
- Add **20** to the end of list **x** and print it again.
- Print now only the second and third value (10, 15) with a single command.
- Replace the first value (5) of **x** with **2**.
- Add **16** and **18** between the values **15** and **20** of **x**.
- Print **x**, it should look like [2, 10, 15, 16, 18, 20].

Task 3

- Print the first 10 Fibonacci numbers. Hint:
- `fib(0) = 0`
- `fib(1) = 1`
- `fib(n) = fib(n-1) + fib(n-2)`

Task 4

- Create two empty lists **even** and **odd**.
- Use a loop to iterate over all numbers between **0** and **19** (both included).
- Within the loop, add all even numbers (0, 2, 4, ...) to list **even** and all odd numbers (1, 3, 5, ...) to list **odd**.

Task 5

- Print all prime numbers between 2 and 20.
- For each number, check if it can be divided without remainder through any number that came before (down to 2). If yes, it is not prime.

Task 6

- Create a method **calculator** with parameters **x1**, **op** and **x2**.
- **x1** and **x2** are numbers, **op** is one of the following math operations: '+', '-', '*', '/'
- The method should return the result of the operation between **x1** and **x2**, e.g. **x1** = 12, **x2** = 5, **op** = '-' will return 7.

Task 7

- Import the **numpy** and **matplotlib.pyplot** module.
- Use them to plot the sine curve between 0 and 20 (in 0.1 steps).

Task 8

- Write a method that returns true when a key is found in a dictionary, false otherwise.

Task 9

- Write a method that returns 'yes' if a given list contains numbers following a linear pattern, otherwise 'no'.
- Example: [1,2,3,4,5] is linear, [2,4,6,8] is also linear, [2,4,7,8] is not.